

Software Requirements Specification

for

Smart Tax Calculator

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version
Smart Tax Calculator Team	11-09-2026	Initial draft prepared from the approved problem statement and feasibility report	1.0
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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the software requirements for the Smart Tax Calculator, Version 1.0. The Smart Tax Calculator is a web-based application that estimates the income-tax liability of an individual taxpayer in India on the basis of the income, deduction and exemption details supplied by the user, and compares that liability under the Old and New Tax Regimes.

This document covers the complete system, including the browser-based user interface, the server-side calculation engine and the supporting database. It does not describe any external system such as the Income Tax Department's e-filing portal; the product neither files returns nor communicates with any government service.

1.2 Document Conventions

The document is prepared in Calibri, with section headings in bold. Functional requirements are uniquely identified using the tag format REQ-n, where n is a running sequence number within the system feature concerned. Each system feature is assigned a priority of High, Medium or Low, and that priority is inherited by every requirement listed beneath it unless stated otherwise.

The word "shall" indicates a mandatory requirement, "should" a desirable one and "may" an optional one. All monetary values are expressed in Indian Rupees (INR). The placeholder "TBD" marks information not yet available; such items are collected in Appendix C.

1.3 Intended Audience and Reading Suggestions

This SRS is intended for the following categories of readers:

- Development team members, who will use the document as the basis for design, coding and integration.
- Course faculty and lab evaluators, who will assess the completeness and consistency of the stated requirements.
- Testers, who will derive test cases from the functional and non-functional requirements.
- End users and domain reviewers, who will verify that the described behaviour matches their expectation of a tax-estimation tool.
- Documentation writers, who will prepare the user manual and on-line help.

The remainder of this document is organised as follows. Section 2 gives a general description of the product, its users and its operating environment. Section 3 specifies the external interfaces. Section 4 details the functional requirements, organised by system feature. Section 5 states the non-functional requirements, and Section 6 covers remaining requirements. Appendices provide a glossary, analysis models and a list of open items.

All readers should begin with Sections 1 and 2. Developers and testers should then read Sections 3, 4 and 5 in full; evaluators and end users may proceed directly from Section 2 to Section 4, referring to Appendix A for terminology.

1.4 Product Scope

The Smart Tax Calculator is a self-contained web application that automates the estimation of income tax for salaried, self-employed and other individual taxpayers. The user enters annual income details, eligible deductions, investments and exemptions, and selects a tax regime. The system computes gross total income, applies the permissible deductions, derives the taxable income, applies the relevant slab rates, and presents an itemised, slab-wise breakdown of the estimated tax liability together with a comparison of the amount payable under the Old and New Tax Regimes.

The principal objectives of the product are to reduce the manual effort and arithmetic errors involved in self-assessment, to make the structure of the tax computation transparent to a non-expert user, and to help the user identify the more suitable regime for their financial situation. The product is developed as an academic project and is intended for informational and educational use; it is not a substitute for advice from a qualified tax professional.

1.5 References

1. IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications, Institute of Electrical and Electronics Engineers, 1998.

2. Smart Tax Calculator — Identification of Problem, Objectives, Scope and Feasibility Analysis (Experiment 2), Software Engineering & Design Principles Lab, KIET Group of Institutions, 2026.
3. Income Tax Department, Government of India — official slab rates, deduction sections and exemption rules. Source: <https://www.incometax.gov.in>
4. Spring Boot Reference Documentation, VMware Tanzu. Source: <https://docs.spring.io/spring-boot/docs/current/reference/html/>
5. MySQL 8.0 Reference Manual, Oracle Corporation. Source: <https://dev.mysql.com/doc/refman/8.0/en/>

2. Overall Description

2.1 Product Perspective

The Smart Tax Calculator is a new, self-contained product. It is not a member of an existing product family and does not replace any deployed system; it automates a process that individual taxpayers presently carry out manually using slab charts, spreadsheet templates or assorted online calculators.

The product follows a conventional three-tier client-server architecture:

- Presentation tier — a browser-based user interface implemented in HTML, CSS and JavaScript, responsible for collecting user input, performing preliminary validation and rendering the tax summary.
- Application tier — a Java Spring Boot application exposing REST endpoints. It contains the calculation engine, which holds the slab definitions, deduction limits and regime rules, and performs the complete tax computation.
- Data tier — a MySQL database that stores the slab and deduction reference data for each supported financial year and, optionally, the calculation records generated by the system.

The tiers communicate over HTTP using JSON-formatted REST messages, and the application tier reaches the database through JDBC. No external system participates in the computation.

2.2 Product Functions

At a high level, the product shall allow the user to:

- Enter annual income details, including salary components and income from other sources.
- Enter eligible deductions, investments and exemptions.
- Select the Old Tax Regime or the New Tax Regime for the computation.
- Obtain the taxable income after the applicable deductions and exemptions have been applied.
- Obtain the estimated tax liability, computed slab by slab, with cess added.
- View a side-by-side comparison of the liability under the Old and New Tax Regimes.
- View a concise, itemised tax summary suitable for a non-expert reader.
- Receive clear validation messages when the data entered is incomplete or invalid.

2.3 User Classes and Characteristics

Four user classes are anticipated. The Salaried Employee and Individual Taxpayer classes are the most important to satisfy, as they represent the majority of expected usage.

User Class	Characteristics	Requirements Relevant to the Class
Individual Taxpayer	Occasional user; limited financial and technical expertise; uses the system once or twice per financial year	All input, calculation, comparison and summary functions; relies heavily on guidance text and validation messages
Salaried Employee	Frequent user during the investment-declaration and filing periods; familiar with salary components and common deduction	Salary-component input, exemption entry, regime comparison, slab-wise breakdown

User Class	Characteristics	Requirements Relevant to the Class
	sections	
Self-Employed User	Enters business or professional income rather than salary; moderate familiarity with tax rules	Other-income input, deduction entry, calculation and summary functions
Project Administrator / Developer	Technical user; maintains slab data, calculation logic and the interface; highest privilege level	Maintenance of slab and deduction reference data, testing, deployment and defect correction

2.4 Operating Environment

The software shall operate in the following environment:

- Client platform — any desktop, laptop or mobile device capable of running a modern web browser (Google Chrome 100 or later, Mozilla Firefox 100 or later, or Microsoft Edge 100 or later), with JavaScript enabled.
- Server platform — Windows 10/11, a Linux distribution, or macOS, with Java Runtime Environment 17 or later installed.
- Application server — the embedded Apache Tomcat container supplied with Spring Boot.
- Database — MySQL 8.x running on the same host as the application server or on an accessible host on the same network.
- Minimum hardware — a dual-core processor (Intel i3 or equivalent), 4 GB RAM (8 GB recommended) and 10 GB of free disk space.

The application must coexist peacefully with other software installed on the host machine and shall not require any modification to the operating-system configuration.

2.5 Design and Implementation Constraints

1. The system shall be implemented using the technology stack approved for the project: HTML, CSS and JavaScript for the frontend; Java with the Spring Boot framework for the backend; and MySQL for persistent storage.
2. Communication between the frontend and the backend shall use REST over HTTP with JSON payloads.
3. The tax slabs, rebate limits, surcharge thresholds, cess rate and deduction ceilings applied by the calculation engine shall be stored as configurable reference data rather than being embedded as literal values in the program code, so that a change in the Finance Act can be accommodated without recompilation.
4. The system shall not connect to any government portal or third-party taxation API; all rules are maintained locally.
5. All monetary computation shall use a fixed-point decimal type rather than floating-point arithmetic, to avoid rounding errors in currency values.
6. The application shall be developed and maintained under Git version control, with the repository hosted on GitHub.
7. Every page presenting a computed result shall carry a visible disclaimer stating that the figures are estimates for informational purposes and do not constitute professional tax advice.
8. Development is constrained by the academic timetable of the semester and by the skill level of a four-member undergraduate team.

2.6 User Documentation

The following user documentation components shall be delivered with the software:

- A user manual in PDF format, describing each screen, the meaning of every input field, and a worked example of a complete calculation.
- On-line contextual help, provided as tooltips and short descriptive text next to input fields whose meaning may not be obvious (for example, deduction sections and exemption categories).

- A README file in the project repository covering installation, database configuration and the procedure for starting the application.
- A frequently-asked-questions page explaining the difference between the two tax regimes and the limitations of the estimate.

2.7 Assumptions and Dependencies

1. It is assumed that the financial details entered by the user are accurate and complete; the system performs no independent verification of the figures supplied.
2. It is assumed that the user has a working internet connection and a supported browser with JavaScript enabled.
3. The correctness of the output depends on the slab rates, deduction limits and rebate rules configured for the selected financial year being current at the time of use.
4. The project depends on the continued free availability of the Spring Boot framework, the MySQL Community Edition database and the associated open-source libraries.
5. It is assumed that tax slabs and deduction rules may change with each annual Union Budget, and that the reference data will require periodic manual updating by the administrator.
6. It is assumed that a single user performs one calculation at a time and that no concurrent multi-user session data needs to be reconciled.

3. External Interface Requirements

3.1 User Interfaces

The user interface shall be a responsive web interface rendered in a standard browser. It shall consist of the following logical screens:

- Home / Introduction screen — a brief description of the tool, the selection of the financial year, and a control to begin a new calculation.
- Income Details screen — fields for basic salary, allowances, other income and gross annual income.
- Deductions and Exemptions screen — fields for eligible deductions, investments and exemptions, each accompanied by its applicable statutory ceiling.
- Regime Selection screen — a control allowing the user to choose the Old Tax Regime, the New Tax Regime, or a comparison of both.
- Results screen — the slab-wise computation, the total estimated liability and the itemised tax summary.
- Comparison screen — a side-by-side presentation of the liability under the two regimes, indicating the lower of the two.

The following interface conventions shall apply to every screen:

- A consistent header carrying the product name and a persistent footer carrying the estimate disclaimer.
- Standard navigation controls labelled Next, Back, Calculate and Reset, positioned consistently across screens.
- A Help control available on every screen.
- Validation errors displayed adjacent to the offending field in a distinct colour, with an explanatory message in plain language rather than a technical error code.
- All currency fields formatted in the Indian numbering convention with the Rupee symbol.

Detailed visual design, including exact screen layouts, colours and typography, shall be documented separately in the user-interface design specification.

3.2 Hardware Interfaces

The product does not interface with any specialised hardware device. It interacts with the client machine only through the standard input and output devices supported by the browser, namely the keyboard, the pointing device and the

display. No printer, scanner, card reader or sensor is required, although the browser's native print function may be used to print the generated tax summary.

On the server side, the application requires only general-purpose computing hardware and a network interface accepting TCP/IP connections. No device drivers beyond those supplied with the operating system are needed.

3.3 Software Interfaces

Component	Name and Version	Nature of the Interface
Database	MySQL 8.x	Accessed by the backend through JDBC using Spring Data JPA. Incoming data: slab definitions, deduction ceilings and stored calculation records. Outgoing data: persisted calculation results.
Backend framework	Spring Boot 3.x (Java 17)	Hosts the calculation service and exposes REST endpoints consumed by the frontend.
Frontend	HTML5, CSS3, JavaScript (ES6)	Sends JSON requests containing income, deduction and regime data; receives JSON responses containing taxable income, slab-wise tax amounts, cess and total liability.
Web browser	Chrome / Firefox / Edge, version 100 or later	Hosts the presentation tier and issues HTTP requests to the application server.
Testing tool	Postman (latest)	Used during development to exercise the REST endpoints directly; not part of the delivered system.

Data shared across the components consists of the calculation request object (income fields, deduction fields, regime selector and financial year) and the calculation response object (gross income, total deductions, taxable income, per-slab tax amounts, rebate, surcharge, cess and total liability). Both are exchanged in JSON and are defined by the REST API documentation maintained with the source code.

3.4 Communications Interfaces

1. All communication between the browser and the application server shall use the HTTP/1.1 protocol; HTTPS shall be used when the application is deployed on a public network.
2. Request and response bodies shall be encoded in JSON with UTF-8 character encoding.
3. The application server shall listen on port 8080 by default; the port shall be configurable.
4. The system shall not send electronic mail, push notifications or any other outbound message on behalf of the user.
5. No data transfer rate is mandated; a standard broadband or mobile data connection is sufficient, as the exchanged payloads are small text messages.
6. Session data, where maintained, shall be synchronised through the standard session-management mechanism of the Spring framework.

4. System Features

The functional requirements of the product are organised below by system feature — that is, by the major services the Smart Tax Calculator provides to its users.

4.1 Income and Financial Details Entry

4.1.1 Description and Priority

This feature allows the user to enter annual income details, including salary components, income from other sources and total gross income, which form the basis of every subsequent computation. Priority: High. (Benefit 9, Penalty 9, Cost 3, Risk 2.)

4.1.2 Stimulus/Response Sequences

Stimulus: The user opens the application and selects the financial year. Response: The system displays the income-details form with all fields blank and the applicable financial year shown in the header.

Stimulus: The user enters a value in an income field and moves focus away from it. Response: The system validates the value and, where valid, updates the running gross-income total displayed on the screen.

Stimulus: The user enters a non-numeric or negative value. Response: The system rejects the entry, retains focus on the field and displays an explanatory validation message.

Stimulus: The user selects Next with all mandatory fields completed. Response: The system stores the income data for the session and displays the deductions screen.

4.1.3 Functional Requirements

REQ-1: The system shall provide input fields for basic salary, allowances, income from house property, income from business or profession and income from other sources.

REQ-2: The system shall accept only non-negative numeric values in all income fields and shall reject alphabetic characters, special characters and negative numbers.

REQ-3: The system shall treat an empty income field as a value of zero, and shall require at least one income field to hold a value greater than zero before the calculation may proceed.

REQ-4: The system shall compute and display the gross total income as the sum of all income components entered.

REQ-5: The system shall display an explanatory message adjacent to any field whose value fails validation, and shall not permit the user to advance to the next step until all such errors are corrected.

REQ-6: The system shall allow the user to clear all entered values and restart the calculation by means of a Reset control.

4.2 Deduction and Exemption Management

4.2.1 Description and Priority

This feature allows the user to enter eligible deductions, investments and exemptions, and applies the statutory ceiling for each category so that the taxable income is derived correctly. Priority: High. (Benefit 9, Penalty 8, Cost 4, Risk 4.)

4.2.2 Stimulus/Response Sequences

Stimulus: The user reaches the deductions screen. Response: The system displays the deduction and exemption categories that are permissible under the selected regime, each showing its applicable ceiling.

Stimulus: The user enters an amount exceeding the statutory ceiling for a category. Response: The system accepts the entry but caps the amount considered in the computation at the ceiling, and informs the user that the excess has not been counted.

Stimulus: The user selects the New Tax Regime. Response: The system disables the deduction categories that are not available under that regime and indicates why they are unavailable.

Stimulus: The user selects Calculate. Response: The system computes the total eligible deductions and derives the taxable income.

4.2.3 Functional Requirements

REQ-7: The system shall provide input fields for the deduction and exemption categories applicable to individual taxpayers, including investment-based deductions, insurance-premium deductions, housing-loan interest and the standard deduction.

REQ-8: The system shall apply the configured statutory maximum to each deduction category and shall disregard any amount entered in excess of that maximum.

REQ-9: The system shall notify the user whenever an entered amount has been capped, stating the ceiling applied.

REQ-10: The system shall enable or disable each deduction category according to the tax regime selected by the user.

REQ-11: The system shall compute the total eligible deduction as the sum of the capped values of all applicable categories.

REQ-12: The system shall compute the taxable income as the gross total income less the total eligible deduction, and shall set the taxable income to zero if the result would otherwise be negative.

4.3 Tax Regime Selection

4.3.1 Description and Priority

This feature allows the user to choose the Old Tax Regime, the New Tax Regime, or a comparison of both, and ensures that the correct slab structure and deduction rules are applied. Priority: High. (Benefit 8, Penalty 8, Cost 2, Risk 2.)

4.3.2 Stimulus/Response Sequences

Stimulus: The user selects a regime. **Response:** The system loads the slab structure and deduction rules associated with that regime for the selected financial year and refreshes the deduction screen accordingly.

Stimulus: The user selects the comparison option. **Response:** The system performs the computation under both regimes using the same input data.

Stimulus: The user changes the regime after entering deduction data. **Response:** The system retains the entered values, re-evaluates which of them are admissible under the newly selected regime, and informs the user of any change.

4.3.3 Functional Requirements

REQ-13: The system shall allow the user to select exactly one of the following: Old Tax Regime, New Tax Regime, or comparison of both.

REQ-14: The system shall default to the comparison option when the user makes no explicit selection.

REQ-15: The system shall load the slab rates, rebate limits and deduction rules corresponding to the selected regime and financial year from the configured reference data.

REQ-16: The system shall preserve all previously entered income and deduction values when the user changes the selected regime.

REQ-17: The system shall display, alongside the selection control, a brief description of the essential difference between the two regimes.

4.4 Tax Computation and Slab-wise Breakdown

4.4.1 Description and Priority

This feature is the core of the product. It computes the estimated tax liability from the taxable income by applying the slab rates of the selected regime, adding the applicable cess and surcharge and subtracting any rebate, and presents the computation step by step. Priority: High. (Benefit 9, Penalty 9, Cost 6, Risk 5.)

4.4.2 Stimulus/Response Sequences

Stimulus: The user selects Calculate with valid income, deduction and regime data. **Response:** The system computes the tax liability and displays the slab-wise breakdown together with the total payable amount.

Stimulus: The computed taxable income falls below the basic exemption limit. **Response:** The system reports a nil tax liability and states the reason.

Stimulus: The taxable income qualifies for a rebate. **Response:** The system applies the rebate, shows it as a separate line in the breakdown and reports the reduced liability.

Stimulus: An internal error occurs during the computation. **Response:** The system displays a non-technical error message, retains the user's input and invites the user to retry.

4.4.3 Functional Requirements

REQ-18: The system shall apply the slab rates of the selected regime progressively, computing the tax attributable to each slab separately.

REQ-19: The system shall display, for every slab, the income range, the applicable rate, the portion of income falling within that slab and the tax computed on it.

REQ-20: The system shall apply the configured rebate where the taxable income is within the prescribed rebate limit, and shall show the rebate as a distinct line item.

REQ-21: The system shall apply the health and education cess at the configured rate to the tax after rebate, and shall display it as a distinct line item.

REQ-22: The system shall apply the applicable surcharge where the taxable income exceeds the configured surcharge threshold.

REQ-23: The system shall round the final tax liability to the nearest rupee and shall display all monetary amounts to two decimal places in the intermediate steps.

REQ-24: The system shall complete and display the computation within three seconds of the Calculate control being selected, under the operating conditions described in Section 2.4.

REQ-25: The system shall report a nil liability, with an accompanying explanation, where the taxable income does not exceed the basic exemption limit.

4.5 Old versus New Regime Comparison

4.5.1 Description and Priority

This feature computes the liability under both tax regimes from a single set of inputs and presents the two results side by side, so that the user can identify the more favourable option. Priority: High. (Benefit 9, Penalty 7, Cost 4, Risk 3.)

4.5.2 Stimulus/Response Sequences

Stimulus: The user selects the comparison option and then Calculate. Response: The system performs two independent computations on the same input data and displays both results in a comparative view.

Stimulus: The two computations produce different liabilities. Response: The system highlights the regime yielding the lower liability and states the amount of the difference.

Stimulus: The two computations produce identical liabilities. Response: The system states that the liability is the same under both regimes.

4.5.3 Functional Requirements

REQ-26: The system shall compute the tax liability under both the Old and the New Tax Regime using a single set of income and deduction inputs.

REQ-27: The system shall apply, in each computation, only those deductions and exemptions that are admissible under the regime concerned.

REQ-28: The system shall present the two results side by side, showing the taxable income, total tax and effective tax rate for each regime.

REQ-29: The system shall identify the regime with the lower liability and shall state the monetary difference between the two.

REQ-30: The system shall accompany the comparison with a statement that the recommendation is indicative only and does not constitute professional tax advice.

4.6 Tax Summary Report

4.6.1 Description and Priority

This feature generates a concise, itemised summary of the computation that a user without a financial background can read and interpret. Priority: Medium. (Benefit 7, Penalty 5, Cost 3, Risk 2.)

4.6.2 Stimulus/Response Sequences

Stimulus: The computation completes successfully. Response: The system generates the tax summary and displays it beneath the slab-wise breakdown.

Stimulus: The user selects Print. Response: The system renders a print-friendly version of the summary through the browser's print function.

Stimulus: The user selects New Calculation. Response: The system clears the session data and returns to the income-details screen.

4.6.3 Functional Requirements

REQ-31: The system shall generate a summary stating the gross total income, total deductions claimed, taxable income, tax before cess, rebate, cess and total estimated liability.

REQ-32: The system shall express the summary in plain language, avoiding unexplained statutory section references and technical abbreviations.

REQ-33: The system shall display the effective tax rate as a percentage of the gross total income.

REQ-34: The system shall provide a print-friendly rendering of the summary.

REQ-35: The system shall display, on the summary, the financial year and the tax regime to which the computation relates.

REQ-36: The system shall display on every summary a disclaimer stating that the figures are estimates for informational purposes only.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

1. The system shall complete a single tax computation and render the result within three seconds of the Calculate control being selected, measured on the minimum hardware configuration stated in Section 2.4.
2. The system shall render a regime comparison, which involves two computations, within five seconds.
3. Any screen of the application shall load within five seconds on a broadband connection of 2 Mbps or better.
4. Field-level input validation shall produce a response within one second of the user leaving the field.
5. The system shall support at least twenty concurrent users without a measurable degradation in response time. This limit reflects the expected academic deployment and is not an architectural ceiling.

5.2 Safety Requirements

1. The system shall display a clearly visible disclaimer on every screen presenting a computed figure, stating that the output is an estimate for informational and educational purposes and is not a substitute for professional tax advice.
2. The system shall not present its output as a final or legally binding tax liability, and shall not be used as the sole basis for filing an income-tax return.
3. The system shall state the financial year and the regime against which a computation was performed, so that an out-of-date result cannot be mistaken for a current one.
4. Where the slab reference data for a selected financial year is missing or incomplete, the system shall refuse to compute and shall inform the user, rather than producing a result from partial data.
5. The system shall preserve the user's entered data when an error occurs, so that no information is lost as a consequence of a failure.

5.3 Security Requirements

1. The system shall not require the user to supply any personally identifiable information, such as a name, Permanent Account Number, address or bank account detail, in order to perform a calculation.
2. The financial data entered shall be retained only for the duration of the session and shall be discarded when the session ends, unless the user explicitly chooses to save the calculation.
3. All communication between the client and the server shall be conducted over HTTPS when the application is deployed on a public network.
4. The system shall validate and sanitise every input on the server side, irrespective of any validation already performed in the browser, so as to prevent injection attacks.
5. Database access shall be performed exclusively through parameterised queries or the object-relational mapping layer; dynamically concatenated SQL statements shall not be used.
6. Administrative functions that modify slab or deduction reference data shall be restricted to authenticated administrator accounts.
7. Internal error details, stack traces and database messages shall never be exposed in the user interface.

5.4 Software Quality Attributes

Attribute	Requirement
Correctness	The tax computed by the system shall match a manual computation performed from the same inputs and the same published slab rates, to within one rupee. Correctness takes precedence over every other quality attribute.
Usability	A first-time user with no background in taxation shall be able to complete a full calculation without external assistance. Ease of use is preferred over ease of learning advanced features.
Reliability	The system shall handle invalid input, missing reference data and internal errors without terminating, and shall remain available for the duration of a user session.
Maintainability	Slab rates, deduction ceilings, rebate limits and the cess rate shall be modifiable through configuration data alone, without any change to the program code.
Portability	The application shall run on Windows, Linux and macOS without modification to the source code, and shall render correctly in the browsers listed in Section 2.4.
Testability	The calculation engine shall be implemented independently of the user interface so that it can be exercised directly by unit tests.
Adaptability	The design shall permit reference data for an additional financial year to be added without structural change to the database schema.
Robustness	No combination of user input shall cause the system to crash, hang or produce an uncaught exception.

5.5 Business Rules

1. Only those deductions and exemptions that are admissible under the tax regime selected by the user shall be applied to the computation.
2. A deduction claimed in excess of its statutory ceiling shall be restricted to that ceiling.
3. Taxable income shall never be recorded as a negative amount; where deductions exceed gross income, the taxable income shall be treated as zero.
4. The rebate shall be applied only where the taxable income falls within the prescribed rebate limit for the selected regime and financial year.
5. Cess shall be computed on the tax remaining after any rebate has been applied, and never on the tax before rebate.
6. Only a user holding the administrator role may create, modify or delete slab, rebate and deduction reference data; ordinary users hold read-only access to that data through the calculation.
7. Every computation shall be associated with exactly one financial year and exactly one tax regime.

6. Other Requirements

6.1 Database Requirements

1. The database shall store the slab definitions for each supported financial year and regime, comprising the lower bound, upper bound and rate of each slab.
2. The database shall store the deduction and exemption categories together with their statutory ceilings and the regimes under which each is admissible.
3. The database shall store the rebate limit, surcharge thresholds and cess rate applicable to each financial year and regime.
4. Where the user chooses to save a calculation, the database shall store the inputs, the computed result and the timestamp, without storing any personally identifiable information.
5. Referential integrity between the slab, regime and financial-year tables shall be enforced by the database.

6.2 Legal Requirements

1. The product is an academic exercise and shall not be presented, distributed or marketed as a professional taxation service.
2. The disclaimer described in Section 5.2 shall be displayed prominently and shall not be removable by the user.
3. The system shall not offer, and shall not purport to offer, legal or financial advice.

6.3 Internationalisation Requirements

1. All monetary values shall be expressed in Indian Rupees and formatted in the Indian numbering system, using the lakh and crore groupings.
2. The user interface shall be presented in English for Version 1.0. The design shall externalise all interface text so that translation into additional languages is possible in a later version.
3. Dates shall be presented in the DD-MM-YYYY format, and financial years shall be identified in the form YYYY-YY.

6.4 Reuse Objectives

1. The calculation engine shall be implemented as an independent service layer so that it may be reused by a future mobile client or exposed as a public API.
2. Input-validation routines shall be implemented as shared utilities usable across all screens of the application.

Appendix A: Glossary

Term	Definition
Assessment Year (AY)	The year following the financial year, in which the income earned during that financial year is assessed to tax.
Cess	An additional levy charged as a percentage of the computed income tax, applied for a designated purpose such as health and education.
Deduction	An amount permitted to be subtracted from gross total income before the tax liability is computed.
Effective Tax Rate	The total tax liability expressed as a percentage of the gross total income.
Exemption	An income component that is excluded from the computation of taxable income.
Financial Year (FY)	The twelve-month period, from 1 April to 31 March, in which income is earned.
Gross Total Income	The aggregate of income from all sources before any deduction is applied.
JDBC	Java Database Connectivity — the standard Java interface for communicating with relational databases.
JSON	JavaScript Object Notation — the text format used for exchanging data between the client and the server.
New Tax Regime	The alternative tax structure offering lower slab rates in exchange for the withdrawal of most deductions and exemptions.
Old Tax Regime	The conventional tax structure permitting a wide range of deductions and exemptions at higher slab rates.
Rebate	A reduction applied to the computed tax where the taxable income falls within a prescribed limit.
REST	Representational State Transfer — the architectural style used for communication between the frontend and the backend.
Slab	A band of income to which a specified rate of tax applies.
SRS	Software Requirements Specification — this document.

Term	Definition
Surcharge	An additional charge levied on the income tax of taxpayers whose income exceeds a prescribed threshold.
Taxable Income	The income remaining after all eligible deductions and exemptions have been applied to the gross total income.
TBD	To Be Determined — a placeholder for information not yet available.

Appendix B: Analysis Models

The following analysis models are to accompany this specification. They are prepared during the design phase and are referenced here for completeness.

Model	Content
B.1 Use Case Diagram	Actors Individual Taxpayer, Salaried Employee, Self-Employed User and Administrator, with the use cases Enter Income Details, Enter Deductions, Select Regime, Calculate Tax, Compare Regimes, View Summary and Maintain Slab Data.
B.2 Level-0 DFD	Context diagram showing the Smart Tax Calculator as a single process, with the User as the sole external entity supplying income, deduction and regime data and receiving the tax summary.
B.3 Level-1 DFD	Decomposition into Validate Input, Compute Taxable Income, Apply Slab Rates, Compare Regimes and Generate Summary, with the Slab Reference data store.
B.4 ER Diagram	Entities FinancialYear, Regime, TaxSlab, DeductionCategory and CalculationRecord, with the relationships between them.
B.5 Class Diagram	Classes TaxCalculationService, RegimeRuleSet, SlabRate, DeductionPolicy and TaxSummary, with the supporting controller and repository classes.
B.6 Activity Diagram	Flow from entry of income details, through validation, deduction entry, regime selection and computation, to presentation of the summary.

Appendix C: To Be Determined List

The following items remain to be determined and are to be tracked to closure before this specification is baselined at Version 2.0.

TBD No.	Description	Section Reference
TBD-1	The financial years for which slab reference data will be supplied in the initial release are not yet fixed.	1.4, 2.7
TBD-2	Whether the system will persist calculation records, or discard all data at the end of the session, is not yet decided.	5.3, 6.1
TBD-3	The mechanism for authenticating administrator accounts has not yet been selected.	5.3, 5.5
TBD-4	Whether the tax summary will be exportable as a PDF in Version 1.0, or only printable through the browser, is not yet decided.	4.6
TBD-5	The treatment of capital-gains income, which is taxed at special rates, has not yet been defined.	4.1
TBD-6	The deployment environment for the final demonstration has not yet been confirmed.	2.4